ELIAS AL HOYEK

Mechatronics & Embedded Systems Engineering Student

Seeking a 4–6 month final-year internship – Available from February 2027

Contact

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Annecy (74000), France
Open to relocation in France

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Portfolio: elias-alhoyek.github.io

Education

2025 – PRESENT

POLYTECH ANNECY – UNIVERSITÉ SAVOIE MONT-BLANC

Master of Science in Mechatronics Engineering

2019 – 2024

AMERICAN UNIVERSITY OF SCIENCE AND TECHNOLOGY, BEIRUT, LEBANON

Bachelor of Science in Computer and Communications Engineering (Biomedical Engineering option)

Technical Skills

- C / C++ programming
- Python and MATLAB (signal processing, data analysis)
- Microcontrollers (ESP32, Arduino)
- Real-time systems (RTOS) and BLE
- Sensors and bioinstrumentation
- Electronics and circuit design
- Technical project management

Soft Skills

- Multidisciplinary teamwork
- Communication and teaching
- Leadership and organization
- Rigor and analytical thinking
- Complex problem solving
- Adaptability and scientific curiosity

Languages

French	Intermediate B1 (self-assessed)
English	Fluent
Arabic	Native

Profile

Mechatronics engineering student with a background in **biomedical engineering**. I bridge **physiology, electronics, embedded programming and mechanics**, translating each domain's constraints to design coherent, integrated devices.

Academic & Research Projects

Adaptive Sampling for Low-Power EMG Devices — Ongoing

Research project – Polytech Annecy

- Developing an amplitude-based adaptive sampling method for EMG signals to extend the battery life of subdermal devices.
- Writing a scientific paper presenting the method and its results.

Fields: signal processing, low-power embedded systems, electromyography

Wireless EMG Acquisition System on ESP32 — 2025 – 2026

Master's project – Polytech Annecy

- Sampled EMG signals on an ESP32 using a multitasking real-time architecture under FreeRTOS.
- Transmitted the data over Bluetooth Low Energy (BLE) to a computer and visualized the received signal.

Technologies: ESP32, FreeRTOS, BLE, C/C++, EMG sensors

Exoskeleton Device for Muscle Atrophy Rehabilitation — June 2024

Final-year project – AUST

- Designed a rehabilitation device that assists grip strength and forearm muscle exercise.
- Integrated control systems, Arduino boards and biomedical instrumentation amplifiers, applying principles of physiology.

Technologies: Arduino, control engineering, bioinstrumentation, electronics

Also: reverse engineering of a calorimeter's analog control board (ongoing).

Professional Experience

Intern – Biomedical Engineering Department — Oct 2023 – Jan 2024

Bellevue Medical Center – Lebanon

- Maintained and repaired biomedical equipment in compliance with safety standards.
- Diagnosed and resolved equipment faults alongside senior technicians in a hospital setting.

Extracurricular Activities

Secretary – IEEE AUST Student Branch · organized events — 2021 – 2023

Mentee – IEEE EMBS Mentoring Program · analyzed data — 2023

Teaching Assistant – AUST Robotics Camp · led workshops — 2023